CS24305 • MODULE 5

Module 5: Networking, Transport & Application Layers

Topics 37 to 52 • IPv4/IPv6, Subnetting, Dijkstra/Bellman-Ford, TCP AIMD & HTTP/3

Module V: Networking, Transport & Application Layers — 16-Topic Master Syllabus Guide

Topic 37: IPv4 Header Format (20-byte base fields, TTL, Protocol, Header Checksum, Fragmentation)

Topic 38: IPv6 Architecture (128-bit addressing, 40-byte base header, Extension Headers vs. IPv4)

Topic 39: Subnetting, Supernetting & Classless Inter-Domain Routing (CIDR Notation & Prefix Masks)

Topic 40: Address Resolution Protocol (ARP Request/Reply, ARP Cache & RARP / DHCP)

Topic 41: Internet Control Message Protocol (ICMP Error Reporting: Ping, Traceroute Mechanics)

Topic 42: Routing Algorithms: Distance Vector Routing (Bellman-Ford & Count-to-Infinity Problem)

Topic 43: Routing Algorithms: Link State Routing (Dijkstra's Shortest Path Algorithm & OSPF)

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Topic 47: TCP Connection Management (3-Way Handshake SYN/SYN-ACK/ACK & 4-Way FIN Teardown)

Topic 48: TCP Flow Control (Sliding Window & Silly Window Syndrome Prevention: Nagle & Clark Algorithms)

Topic 49: TCP Congestion Control (AIMD, Slow Start, Congestion Avoidance, Fast Retransmit, Fast Recovery)

Topic 50: Domain Name System (DNS Hierarchical Namespace, Root Servers, Iterative vs Recursive Queries)

Topic 51: Electronic Mail Architecture (SMTP, POP3, IMAP4, MIME & Message Transfer Agents)

Topic 52: Web Protocols: HTTP/1.0, HTTP/1.1 (Persistent Connections & Pipelining), HTTP/2 & HTTP/3 QUIC

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Topic 37 & 39: IPv4 Header Architecture, Subnetting & CIDR Math

IPv4 Field	Field Size	Core Routing Function & Protocols
Version	4 Bits	Specifies IP version (`0100` = IPv4, `0110` = IPv6).
IHL (Header Length)	4 Bits	Header length in 32-bit words (Minimum value = 5 $\implies$ 20 bytes; Maximum = 15 $\implies$ 60 bytes).
Total Length	16 Bits	Total datagram size (Header + Data) in bytes (Max $65,535 \text{ bytes}$).
Identification	16 Bits	Unique integer tagging all fragments belonging to same original datagram.
Flags	3 Bits	Bit 0 (Reserved); Bit 1 (DF = Don't Fragment); Bit 2 (MF = More Fragments).
Fragment Offset	13 Bits	Offset of fragment relative to original unfragmented payload in **8-byte blocks**.
TTL (Time to Live)	8 Bits	Hop counter decremented by 1 at every router; discarded at 0 with ICMP Time Exceeded.
Protocol	8 Bits	Specifies Layer 4 payload: ICMP = 1, IGMP = 2, TCP = 6, UDP = 17, OSPF = 89.
Header Checksum	16 Bits	1's complement checksum covering ONLY the IP header (recomputed at every hop).
Source & Dest IP	32 Bits each	Origin and final destination IPv4 logical addresses.

🏛 Step-by-Step Solved Problem 1: IPv4 Fragmentation Calculation

An IPv4 packet with Total Length = 4020 bytes (20 bytes IP header + 4000 bytes data) travels across an Ethernet link with $\text{MTU} = 1500 \text{ bytes}$. Calculate the fields for all resulting fragments.

Solution: Max payload per fragment = $1500 - 20 = 1480 \text{ bytes}$ (divisible by 8: $1480/8 = 185$).

- **Fragment 1:** Total Length = 1500 B (20 header + 1480 data), $\text{MF} = 1, \text{Offset} = 0/8 = \mathbf{0}$. (Carries bytes $0 \ldots 1479$).
- **Fragment 2:** Total Length = 1500 B (20 header + 1480 data), $\text{MF} = 1, \text{Offset} = 1480/8 = \mathbf{185}$. (Carries bytes $1480 \ldots 2959$).
- **Fragment 3:** Total Length = 1060 B (20 header + 1040 data), $\text{MF} = 0, \text{Offset} = 2960/8 = \mathbf{370}$. (Carries bytes $2960 \ldots 3999$).

🏛 Step-by-Step Solved Problem 2: CIDR Subnetting & Address Allocation

An ISP assigns block `198.51.100.0/24` to an organization needing 4 equal subnets. Find: (1) Subnet mask, (2) Usable hosts per subnet, (3) Subnet network addresses, and (4) Broadcast addresses.

Solution:

- To create 4 subnets: Borrow $k = \log_2(4) = 2$ bits $\implies$ New prefix $= /24 + 2 = /\mathbf{26}$.
- **Subnet Mask:** $255.255.255.192$ (Binary: `11111111.11111111.11111111.11000000`).
- **Usable Hosts per Subnet:** $2^{(32-26)} - 2 = 2^6 - 2 = 64 - 2 = \mathbf{62} \text{ hosts}$.
- **Subnet 1:** Network: `198.51.100.0/26`, Range: `.1 - .62`, Broadcast: `198.51.100.63`
- **Subnet 2:** Network: `198.51.100.64/26`, Range: `.65 - .126`, Broadcast: `198.51.100.127`
- **Subnet 3:** Network: `198.51.100.128/26`, Range: `.129 - .190`, Broadcast: `198.51.100.191`
- **Subnet 4:** Network: `198.51.100.192/26`, Range: `.193 - .254`, Broadcast: `198.51.100.255`

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Topic 42 & 43: Routing Algorithms: Distance Vector vs. Link State (Dijkstra)

Dimension	Distance Vector Routing (Bellman-Ford / RIP)	Link State Routing (Dijkstra / OSPF)
Knowledge of Topology	Routers know only distance vectors of immediate direct neighbors ("Routing by rumor").	Every router possesses complete global topological graph of entire Autonomous System.
Algorithm	Bellman-Ford Equation: $D_x(y) = \min_v\{c(x,v) + D_v(y)\}$.	Dijkstra's Shortest Path Tree algorithm.
Convergence Speed	Slow convergence; suffers from **Count-to-Infinity** problem on link failures.	**Fast Convergence:** Link State Advertisements (LSAs) flood updates instantly.
Loop Prevention	Split Horizon, Poison Reverse, Maximum Hop Count (= 15 in RIP).	Zero routing loops (Dijkstra guarantees unique loop-free shortest path tree).

Topic 46, 47 & 49: TCP Architecture, Handshakes & Congestion Control

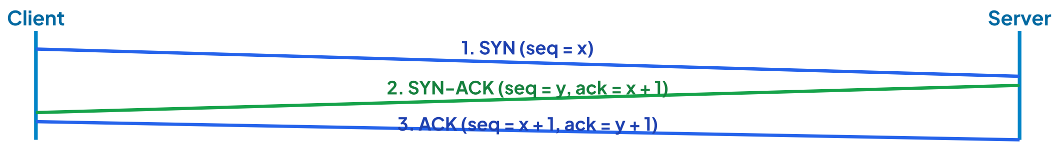

Figure 5.1: TCP 3-Way Handshake Connection Establishment Protocol

The 4 Phases of TCP Congestion Control (AIMD): 1. **Slow Start:** Starts with $\mathrm{cwnd} = 1\ \mathrm{MSS}$. Doubles cwnd every RTT ($\mathrm{cwnd} \leftarrow \mathrm{cwnd} \times 2$) exponentially until reaching $\mathrm{ssthresh}$. 2. **Congestion Avoidance:** Linear growth: increases cwnd by $1\ \mathrm{MSS}$ per RTT ($\mathrm{cwnd} \leftarrow \mathrm{cwnd} + 1$) until packet loss occurs. 3. **Timeout Reaction (TCP Tahoe):** Sets $\mathrm{ssthresh} \leftarrow \mathrm{cwnd}/2$, resets $\mathrm{cwnd} \leftarrow 1\ \mathrm{MSS}$, re-enters Slow Start. 4. **3 Duplicate ACKs (TCP Reno):** Fast Retransmit triggers immediate retransmission; sets $\mathrm{ssthresh} \leftarrow \mathrm{cwnd}/2$, $\mathrm{cwnd} \leftarrow \mathrm{ssthresh} + 3\ \mathrm{MSS}$, enters Fast Recovery!

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Topic 50, 51 & 52: Application Layer Protocols (DNS, Email & HTTP)

Protocol	Port & Transport	Architecture & Function	Key Evolution Highlights
DNS	Port 53 (UDP primary, TCP for zone transfer)	Hierarchical distributed database: Root → TLD (.com, .edu) → Authoritative servers.	Iterative vs. Recursive resolution; DNSSEC security records.
SMTP	Port 25 / 587 (TCP)	Push protocol for sending email between Message Transfer Agents (MTAs).	MIME extensions support attachments, audio, HTML formatting.
POP3 / IMAP4	POP3: Port 110; IMAP: Port 143 (TCP)	Pull protocols for retrieving email from mail servers to local user mail clients.	POP3 downloads and deletes; IMAP syncs folders dynamically across multiple devices.
HTTP/1.1	Port 80 (TCP)	Persistent TCP connections (`Keep-Alive`) & pipelining over single connection.	Suffers from Head-of-Line (HoL) blocking at application level.
HTTP/2	Port 443 (TLS/TCP)	Binary framing layer; multiplexes multiple independent request/response streams over 1 TCP socket.	HPACK header compression, Server Push; TCP-level packet drop still stalls all streams.
HTTP/3 (QUIC)	Port 443 (UDP)	Replaces TCP with QUIC over UDP: combines TLS 1.3 handshake and transport in single round-trip!	**Zero Head-of-Line Blocking:** Packet loss in one stream never stalls other streams!

🧠 M5 Active Recall & 10-Question University Exam Master Bank

Q1. Draw the complete IPv4 Header format and explain the function of TTL, Header Checksum, and Fragmentation fields. (10 Marks)

• **TTL (8 bits):** Hop counter decremented at each router. Prevents undeliverable packets from looping infinitely.
• **Header Checksum (16 bits):** 1's complement sum covering header only. Verified at every hop.
• **Identification (16 bits):** Tags fragments of same datagram.
• **Flags (3 bits):** DF (Don't Fragment), MF (More Fragments).
• **Fragment Offset (13 bits):** Measures data offset in 8-byte blocks relative to unfragmented payload.

Q2. Explain TCP 3-Way Handshake Connection Establishment and 4-Way Connection Termination with sequence diagrams. (10 Marks)

• **3-Way Establishment:** (1) Client → Server: `SYN (seq=x)`; (2) Server → Client: `SYN-ACK (seq=y, ack=x+1)`; (3) Client → Server: `ACK (seq=x+1, ack=y+1)`.
• **4-Way Teardown:** (1) Client → Server: `FIN`; (2) Server → Client: `ACK` (half-close); (3) Server → Client: `FIN`; (4) Client → Server: `ACK` (Client waits $2\ \mathrm{MSL}$ in `TIME_WAIT` before closing).

Q3. Explain TCP Congestion Control mechanisms: Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery. (10 Marks)

• **Slow Start:** Exponential growth: $\text{cwnd} \leftarrow \text{cwnd} \times 2$ per RTT until reaching ssthresh.
• **Congestion Avoidance:** Additive linear growth: $\text{cwnd} \leftarrow \text{cwnd} + 1 \text{ MSS}$ per RTT.
• **Fast Retransmit:** 3 duplicate ACKs trigger immediate retransmission without waiting for RTO timeout.
• **Fast Recovery:** Multiplicative decrease: $\text{ssthresh} = \text{cwnd}/2$, sets $\text{cwnd} = \text{ssthresh} + 3 \text{ MSS}$, continues congestion avoidance.

Q4. Compare Distance Vector (Bellman-Ford) and Link State (Dijkstra) routing algorithms. How is Count-to-Infinity solved? (8 Marks)

• **Distance Vector:** Periodic routing table exchange with neighbors only; slow convergence; suffers Count-to-Infinity. Solved by **Split Horizon** (do not advertise route back to neighbor who supplied it) and **Poison Reverse** (advertise cost as ∞).
• **Link State:** Floods Link State Advertisements (LSAs) to all routers; builds global topology map; runs Dijkstra's algorithm; zero routing loops!

Q5. Compare HTTP/1.1, HTTP/2, and HTTP/3 QUIC across multiplexing, transport layer, and Head-of-Line blocking. (8 Marks)

• **HTTP/1.1 (TCP):** Sequential requests per socket; application Head-of-Line (HoL) blocking.
• **HTTP/2 (TCP):** Binary multiplexed streams over single TCP connection; single packet drop stalls all streams (TCP HoL blocking).
• **HTTP/3 (QUIC/UDP):** Native multiplexed streams over UDP; packet drop in stream 1 never delays stream 2 (Zero HoL blocking!).

Topic 52.2: Comprehensive Network Layer & Transport Layer Solved Numericals

🏛 Solved Numerical 1: Dijkstra's Link State Shortest Path Algorithm Trace

Find the shortest paths from source router A to all nodes in network graph:

- Edges: $A - B : 2,\ A - C : 5,\ B - C : 2,\ B - D : 4,\ C - D : 1,\ C - E : 5,\ D - E : 1$

Execution Trace Table:

Step	Visited Set N'	$D(B), p(B)$	$D(C), p(C)$	$D(D), p(D)$	$D(E), p(E)$
0	$\{A\}$	$2, A$	$5, A$	∞	∞
1	$\{A, B\}$	**2, A**	$4, B$	$6, B$	∞
2	$\{A, B, C\}$	2, A	**4, B**	$5, C$	$9, C$
3	$\{A, B, C, D\}$	2, A	4, B	**5, C**	$6, D$
4	$\{A, B, C, D, E\}$	2, A	4, B	5, C	**6, D**

Shortest Paths from A: to $B = 2$, to $C = 4$ ($A \to B \to C$), to $D = 5$ ($A \to B \to C \to D$), to $E = 6$ ($A \to B \to C \to D \to E$).

🏛 Solved Numerical 2: TCP AIMD Congestion Window Evolution

A TCP Reno connection has $\text{ssthresh} = 16\ \text{MSS}$. It starts in Slow Start ($\text{cwnd} = 1$). Trace cwnd over successive RTTs until a timeout occurs at $\text{cwnd} = 20$, followed by 3 duplicate ACKs at $\text{cwnd} = 12$:

1. **RTT 1–5 (Slow Start):** cwnd doubles: $1 \to 2 \to 4 \to 8 \to 16\ \text{MSS}$ (Reaches ssthresh).
2. **RTT 6–9 (Congestion Avoidance):** Linear growth: $17 \to 18 \to 19 \to 20\ \text{MSS}$.
3. **Timeout at $\text{cwnd} = 20$:** Sets $\text{ssthresh} \leftarrow 20/2 = 10\ \text{MSS}$, resets $\text{cwnd} \leftarrow 1\ \text{MSS}$, re-enters Slow Start.
4. **RTT 10–14:** Doubles $1 \to 2 \to 4 \to 8 \to 10$ (ssthresh reached), then linear: $11 \to 12\ \text{MSS}$.
5. **3 Duplicate ACKs at $\text{cwnd} = 12$:** Fast Retransmit! Sets $\text{ssthresh} \leftarrow 12/2 = 6\ \text{MSS}$, sets $\text{cwnd} \leftarrow 6 + 3 = 9\ \text{MSS}$ in Fast Recovery!

Topic 52.3: Comprehensive Internetworking, BGP & Transport Internals

In global autonomous internetworking, routing is divided into **Intra-Domain Interior Gateway Protocols (IGP: OSPF, IS-IS)** and **Inter-Domain Exterior Gateway Protocols (EGP: BGP-4)**:

Routing Protocol	Algorithm Family	Underlying Transport Protocol	Routing Metric & Domain
RIPv2	Distance Vector (Bellman-Ford)	UDP (Port 520)	Hop Count (Max = 15 hops; 16 = ∞); Small LANs.
OSPFv2 / OSPFv3	Link State (Dijkstra)	Raw IP (Protocol 89)	Cost = $\frac{10^8}{\text{Bandwidth (bps)}}$; Large Enterprise / Campus.
BGP-4	Path Vector	TCP (Port 179)	AS-Path list, Policy-based routing, Local Preference; Global Internet Core.

🏛 Complete Step-by-Step Distance Vector Routing & Count-to-Infinity Trace

Consider 3 linear routers $A \leftrightarrow B \leftrightarrow C$ with unit link costs (1):

- Initial state: B reaches C in 1 hop ($D_B(C) = 1$); A reaches C via B in 2 hops ($D_A(C) = 2$).
- **Link $B - C$ Fails:** B detects failure. But before B advertises ∞, A sends its periodic update stating: *"I can reach C with cost 2"*.
- B incorrectly concludes: *"I can reach C via A with cost $2 + 1 = 3$"*. B updates $D_B(C) = 3$.
- On next iteration, A learns B's cost is 3, updating $D_A(C) = 3 + 1 = 4$.
- B updates $D_B(C) = 5 \implies A$ updates $D_A(C) = 6 \ldots$ **Counting to Infinity!**

Remedies: (1) *Split Horizon:* A never advertises its route to C back to B. (2) *Poison Reverse:* A advertises $D_A(C) = \infty$ to B.

🏛 Step-by-Step Silly Window Syndrome Prevention: Nagle vs. Clark Algorithms

Silly Window Syndrome occurs when data is exchanged in tiny 1-byte segments (e.g. typing single characters in Telnet), causing a 40-byte TCP/IP header overhead for every single byte of payload:

- **Sender-Side Solution (Nagle's Algorithm):** The sender transmits the first small packet immediately. While waiting for ACK, the sender buffers all subsequent outgoing data into a full Maximum Segment Size (MSS) block before transmitting the next segment.
- **Receiver-Side Solution (Clark's Solution):** The receiver prevents sending tiny window update advertisements. It advertises a window size of 0 until its internal buffer has space for either a full MSS or half the receiver buffer capacity!

Topic 52.4: Master University Solved Examination Numerical Bank

🏛 Solved Numerical 3: Hierarchical CIDR Route Aggregation & Supernetting

A regional ISP router receives routing updates for four contiguous IP address blocks:

- Block 1: `200.16.0.0/24` (Binary: `11001000.00010000.00000000.00000000`)
- Block 2: `200.16.1.0/24` (Binary: `11001000.00010000.00000001.00000000`)
- Block 3: `200.16.2.0/24` (Binary: `11001000.00010000.00000010.00000000`)
- Block 4: `200.16.3.0/24` (Binary: `11001000.00010000.00000011.00000000`)

Route Aggregation (Supernetting) Solution:

Comparing the 3rd octets in binary: `00000000`, `00000001`, `00000010`, `00000011`. All 4 blocks share the identical first 22 bits (`11001000.00010000.000000xx`).

$$\text{Single Aggregated Supernet Route} = \mathbf{200.16.0.0/22} \quad (\text{Subnet Mask: } \mathbf{255.255.252.0})$$

Benefit: Replaces 4 independent routing table entries with 1 consolidated entry, reducing router BGP memory table size by 75%!

🏛 Solved Numerical 4: TCP Maximum Throughput & Mathis Equation

A cross-continental 10 Gbps TCP link has an RTT of 100 ms and standard MSS of 1460 bytes. If the packet loss rate is $p = 0.0001$ (0.01%), calculate the maximum achievable TCP throughput according to the **Mathis Formula**.

Solution (Mathis Formula):

$$\text{Throughput} \leq \frac{\text{MSS}}{\text{RTT}} \times \frac{\mathbf{1.22}}{\sqrt{\mathbf{p}}}$$

$$\text{Throughput} \leq \frac{1460 \times 8\text{ bits}}{0.100\text{ s}} \times \frac{1.22}{\sqrt{0.0001}} = 116,800 \times \frac{1.22}{0.01} = 116,800 \times 122 = \mathbf{14,249,600}\text{ bps} \approx \mathbf{14.25}\text{ Mbps}$$

Critical Engineering Insight: Even on a 10 Gbps physical pipe, a tiny 0.01% packet drop throttles standard TCP Reno throughput down to 14.25 Mbps due to aggressive AIMD window halving!

Topic 52.5: Comprehensive Transport State Machines & Variable Length Subnetting (VLSM)

🏛 Solved Numerical 5: Complete Variable-Length Subnet Masking (VLSM) Design

An enterprise organization is assigned address block `172.16.0.0/16`. Design an optimal VLSM subnetting plan for 4 distinct departments:

- Department A: Requires 4000 host addresses
- Department B: Requires 2000 host addresses
- Department C: Requires 1000 host addresses
- Department D: Requires 500 host addresses

VLSM Allocation Plan:

Department	Hosts Needed	Bits Needed ($2^h - 2 \geq N$)	Prefix Mask	Assigned Network Address	Usable Host Range
Dept A	4000	$h = 12\ (4094)$	$/20\ (255.255.240.0)$	`172.16.0.0/20`	`172.16.0.1 - 172.16.15.254`
Dept B	2000	$h = 11\ (2046)$	$/21\ (255.255.248.0)$	`172.16.16.0/21`	`172.16.16.1 - 172.16.23.254`
Dept C	1000	$h = 10\ (1022)$	$/22\ (255.255.252.0)$	`172.16.24.0/22`	`172.16.24.1 - 172.16.27.254`
Dept D	500	$h = 9\ (510)$	$/23\ (255.255.254.0)$	`172.16.28.0/23`	`172.16.28.1 - 172.16.29.254`

🏛 Step-by-Step TCP Jacobson's RTO Calculation Algorithm

Given initial $\text{SRTT} = 100\ \text{ms}$, $\text{RTTVAR} = 10\ \text{ms}$, and $\alpha = 0.125$, $\beta = 0.25$. If a new sample RTT measurement $R = 140\ \text{ms}$ arrives, compute the updated Retransmission Timeout (RTO):

$$\text{Error } E = R - \text{SRTT} = 140 - 100 = +\mathbf{40}\ \text{ms}$$

$$\text{Updated SRTT} \leftarrow \text{SRTT} + \alpha \times E = 100 + 0.125(40) = 100 + 5 = \mathbf{105}\ \text{ms}$$

$$\text{Updated RTTVAR} \leftarrow \text{RTTVAR} + \beta \times (|E| - \text{RTTVAR}) = 10 + 0.25(40 - 10) = 10 + 7.5 = \mathbf{17.5}\ \text{ms}$$

$$\text{New RTO} = \text{SRTT} + \mathbf{4} \times \text{RTTVAR} = \mathbf{105} + \mathbf{4(17.5)} = \mathbf{105} + \mathbf{70} = \mathbf{175}\ \text{ms}$$

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🏛️ Solved Numerical 6: Bellman-Ford Distance Vector Calculation with Routing Table Updates

Given 4 routers A, B, C, D. Router B has direct neighbor costs: $c(B, A) = 3, c(B, C) = 1, c(B, D) = 6$. B receives distance vectors from A and C:

- From A: $D_A = (A : 0, B : 3, C : 4, D : 2)$
- From C: $D_C = (A : 4, B : 1, C : 0, D : 3)$

Update B's routing table for destination D using Bellman-Ford: $D_B(D) = \min\{c(B, A) + D_A(D),\ c(B, C) + D_C(D),\ c(B, D)\}$.

$$\text{Via } A : 3 + 2 = 5$$

$$\text{Via } C : 1 + 3 = 4$$

$$\text{Direct } D : 6$$

$$\mathbf{D_B(D) = \min(5, 4, 6) = 4} \quad (\text{Next Hop: Router } \mathbf{C})$$

Q6. Compare IPv4 and IPv6 across address space, header size, fragmentation, checksum, and security. (10 Marks)

• **Address Space:** IPv4 has 32-bit addresses (4.3×10^9); IPv6 has 128-bit addresses (3.4×10^{38}).
• **Header Size:** IPv4 is variable ($20\text{–}60$ B); IPv6 has a fixed 40-byte base header with chained Extension Headers.
• **Fragmentation:** IPv4 routers perform fragmentation; in IPv6, ONLY the source host fragments packets (Path MTU Discovery).
• **Header Checksum:** IPv4 recomputes checksum at every hop; IPv6 eliminates header checksum to maximize router forwarding speed.
• **Security:** IPsec is optional add-on in IPv4; IPsec support is built natively into the core IPv6 protocol specification!

Q7. Explain Address Resolution Protocol (ARP), ARP Request (Broadcast), ARP Reply (Unicast), and ARP Cache Poisoning. (8 Marks)

• **ARP:** Maps a known Layer 3 logical IP address to an unknown Layer 2 physical MAC address on the local subnet.
• **ARP Request:** Broadcast frame (`FF:FF:FF:FF:FF:FF`) asking: *"Who has IP X.X.X.X? Tell MAC Y."*
• **ARP Reply:** Unicast frame from the target host stating: *"I have IP X.X.X.X, my MAC is Z."*
• **ARP Poisoning (Spoofing):** An attacker sends gratuitous fake ARP replies associating the gateway's IP with the attacker's MAC, enabling Man-in-the-Middle (MitM) packet sniffing!

Topic 52.6: Advanced TCP State Transition Machine & Congestion Control Math

Figure 5.2: The Classical TCP Client Connection Lifetime State Machine

🏛️ Step-by-Step Solved Problem: Variable Length Subnet Masking (VLSM) Enterprise Plan

An enterprise is allocated IP address block `10.0.0.0/16`. Subnet the block for: (1) Head Office: 10,000 hosts, (2) Branch 1: 5,000 hosts, (3) Branch 2: 2,000 hosts, (4) Branch 3: 1,000 hosts, (5) Five WAN point-to-point links (2 hosts each).

- **Head Office (10,000 hosts):** $2^{14} - 2 = 16,382 \geq 10,000 \implies$ Mask: $/18\ (255.255.192.0) \implies$ $\mathbf{10.0.0.0/18}$ (`.0.1` to `.63.254`).
- **Branch 1 (5,000 hosts):** $2^{13} - 2 = 8,190 \geq 5,000 \implies$ Mask: $/19\ (255.255.224.0) \implies$ $\mathbf{10.0.64.0/19}$ (`.64.1` to `.95.254`).
- **Branch 2 (2,000 hosts):** $2^{11} - 2 = 2,046 \geq 2,000 \implies$ Mask: $/21\ (255.255.248.0) \implies$ $\mathbf{10.0.96.0/21}$ (`.96.1` to `.103.254`).
- **Branch 3 (1,000 hosts):** $2^{10} - 2 = 1,022 \geq 1,000 \implies$ Mask: $/22\ (255.255.252.0) \implies$ $\mathbf{10.0.104.0/22}$ (`.104.1` to `.107.254`).
- **5 WAN Links (2 hosts each):** Mask $/30\ (255.255.255.252) \implies$ `10.0.108.0/30`, `10.0.108.4/30`, `10.0.108.8/30`, `10.0.108.12/30`, `10.0.108.16/30`.

Efficiency: 100% zero address collisions with over $40,000$ IP addresses preserved for future organizational growth!

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